



Integrated Shared Parking Management at Religious Tourism Destinations: User-Adaptive Capacity Allocation, Vehicle Circulation Control, and Digital Parking Guidance for Peak-Hour Congestion Mitigation

Alfia Magfirona^{1*}, Nurul Hidayati¹, Gunawan Ariyanto², Anggia Dyah Nirmala Dewi¹, Fadhilla Tri Nugrahaini³, Muhammad Ali Rofiq⁴

¹ Civil Engineering Department, Universitas Muhammadiyah Surakarta, 57162 Surakarta, Indonesia

² Informatics Engineering Department, Universitas Muhammadiyah Surakarta, 57162 Surakarta, Indonesia

³ Architecture Department, Universitas Muhammadiyah Surakarta, 57162 Surakarta, Indonesia

⁴ Civil Engineering Department, National Central University, 320317 Taoyuan City, Taiwan, China

* Correspondence: Alfia Magfirona (am389@ums.ac.id)

Received: 01-05-2026

Revised: 05-18-2026

Accepted: 05-25-2026

Citation: A. Magfirona, N. Hidayati, G. Ariyanto, A. D. N. Dewi, F. T. Nugrahaini, and M. A. Rofiq, "Integrated shared parking management at religious tourism destinations: User-adaptive capacity allocation, vehicle circulation control, and digital parking guidance for peak-hour congestion mitigation," *Int. J. Transp. Dev. Integr.*, vol. 10, no. 2, pp. 572–586, 2026. <https://doi.org/10.56578/ijtdi100215>.



© 2026 by the author(s). Licensee Acadlore Publishing Services Limited, Hong Kong. This article can be downloaded for free, and reused and quoted with a citation of the original published version, under the CC BY 4.0 license.

Abstract: The rapid growth of religious tourism has intensified the demand for supporting transport infrastructure, particularly an efficient shared parking system integrated with on-site traffic circulation and pedestrian flow management at sacred sites. This study defines the shared parking scheme as the temporal and spatial allocation of a common facility among buses (for organized pilgrim tours) and passenger cars (for individual visitors), managed by the mosque authority across distinct worship-time windows. Three research questions are addressed: (i) whether visitor groups differ in acceptable walking distance to parking; (ii) whether a digital parking guidance system is suitable across age cohorts; and (iii) how vehicle type influences parking capacity planning. A questionnaire survey was administered to 505 respondents at the Sheikh Zayed Grand Mosque in Surakarta, Indonesia. The Pearson Chi-Square Test examined associations between categorical variables; where more than 20% of cells had expected frequencies below five, Fisher's Exact Test with Monte Carlo approximation was applied, and Cramer's V was reported as the effect-size measure. Age was significantly associated with nearly all parking-preference variables ($p < 0.01$), with the 17–32-year cohort showing higher receptivity to digital parking information systems. Vehicle type exhibited significant associations with five of six preference variables ($p < 0.05$) and the largest mean Cramer's V, indicating the most consistent though not causal demographic correlate. Travel purpose was significantly associated with visiting duration ($p = 0.007$) and acceptable walking distance ($p = 0.043$). Findings yield four operational recommendations: (i) segregated bus and passenger-car zones with dedicated bus-reservation slots; (ii) tiered short-stay/long-stay zoning aligned with prayer-time peaks; (iii) age-differentiated wayfinding combining digital guidance and on-site human assistance; and (iv) temporary traffic control during peak worship hours.

Keywords: Shared parking; Capacity allocation; Vehicle circulation; Digital guidance; Religious tourism

1 Introduction

The growth of religious tourism has increased significantly in recent decades, driving the need for adequate transport infrastructure to accommodate the surge in visitor numbers. One important aspect of this infrastructure is the provision of efficient and effective parking facilities to support the smooth flow of tourism-related road traffic [1]. Visitor parking preferences are influenced by a variety of factors, including demographic characteristics such as gender, age, income, travel purpose, and vehicle type, which shape their habits and comfort levels in choosing parking locations [2]. Understanding how these demographic factors influence parking preferences can help in the planning and management of parking facilities that are more adaptive and responsive to user characteristics [3, 4].

From an integrated transport-management perspective, religious tourism destinations exhibit travel-demand

patterns that differ markedly from those of typical urban or commercial sites. Fixed worship schedules, the five daily prayers, and the Friday congregational prayer concentrate visitor arrivals into narrow time windows, while the simultaneous presence of motorcycle riders, passenger-car users, group-tour buses, and minibuses (locally known as *elf*) generates a heterogeneous vehicle mix that conventional single-user parking facilities are not designed to accommodate. These conditions produce sharp peaks in parking demand, complex vehicle circulation patterns, and frequent pedestrian–vehicle conflicts at the destination entrance plaza operational features that fall squarely within the domain of road traffic management and transport-system integration. The shared parking concept, in which a single facility serves multiple user groups with non-coincident peak demands, is therefore particularly relevant to religious tourism, offering a basis for adaptive capacity allocation, vehicle circulation control, and digital parking guidance as integrated measures for sustainable transport planning at sacred destinations.

Several studies have examined the relationship between demographic factors and parking preferences. Egan et al. [5] reported significant associations between demographic factors and parking preferences using Principal Component Analysis and Cluster Analysis on 574 respondents, finding that gender and education level influenced parking choices. Zong et al. [6], using Structural Equation Modeling (SEM) in Beijing, found that monthly income was significantly associated with parking decisions, with higher-income users more likely to choose off-street parking and to park for shorter durations. Widodo et al. [7] in Sidoarjo applied a logistic regression model and found that age, gender, parking frequency, and parking duration were significantly associated with online parking choice, with women and younger users showing a higher propensity to adopt the service. Zhang et al. [8] used a Hierarchical Regression Model and showed that age, education, presence of children, and driving experience were significantly associated with the intention to adopt Smart Parking Systems.

In the tourism context, Dewi et al. [9] found that age and occupation influenced tourists' mode-choice preferences at Matahari Terbit Beach, Bali. Wirasutama et al. [10] emphasised the importance of parking-capacity analysis to avoid roadside-parking-induced congestion at the Penglipuran Tourism Village, Bali. However, most existing studies still focus on general tourism destinations, while religious tourism destinations exhibit distinct visitor-behaviour patterns driven by spiritual motivations, communal religious activities, and a heightened need for accessibility and pedestrian comfort [11]. Although various studies have addressed the relationship between demographic factors and parking preferences, most have been conducted in urban areas, shopping centres, or general tourism destinations [12]. Research that specifically integrates demographic-based parking-preference analysis with the operational design of shared parking systems at religious tourism destinations, including bus/car zoning, digital versus on-site parking guidance, and peak-period traffic control, remains scarce.

To address this gap, the present study is structured around three research questions that frame the manuscript as a transport planning and parking management investigation rather than a general questionnaire-based statistical exercise. The first question (RQ1) examines whether demographic subgroups of religious-tourism visitors differ significantly in their acceptable walking distance from parking facilities to the mosque entrance, since walking-distance tolerance is a fundamental input to the spatial design of parking zones and pedestrian-circulation routes. The second question (RQ2) investigates whether a digital shared-parking information system is equally suitable across age cohorts, or whether its perceived ease of use varies with age and vehicle type, a distinction that determines the design of hybrid digital-and-on-site parking guidance. The third question (RQ3) explores how vehicle type (motorcycle, passenger car, bus, and minibus) shapes parking-capacity requirements and visiting-duration patterns at a religious tourism destination, and what these patterns imply for bus–car spatial segregation, bus-reservation protocols, and peak-hour vehicle circulation control.

These questions are answered by analysing the association between five demographic factors (gender, age, income, travel purpose, and vehicle type) and six parking-preference variables, using the Pearson Chi-Square Test, Fisher's Exact Test with Monte Carlo approximation for larger contingency tables and sparse cells, and Cramer's V as the effect-size measure. The novelty of the study lies in the explicit translation of demographic-based statistical associations into operational transport-management deliverables for religious tourism destinations, including user-adaptive parking-capacity allocation, bus/car zoning, vehicle circulation control, and the design of digital and on-site parking guidance for different user groups, repositioning the contribution from a behavioural survey toward an integrated parking-management framework.

2 Literature Review

The growing complexity of urban and tourism-destination mobility has positioned parking management as a critical area of transportation research. Three interconnected research streams emerge in the recent literature: parking choice behavior and preferences, shared parking and smart parking systems, and transport management at religious tourism destinations.

2.1 Parking Choice Behavior and Preferences

Empirical studies consistently identify parking cost and walking distance as dominant correlates of driver behaviour, with higher fees deterring users and shorter distances to destinations being universally preferred [1, 2]. Preferences between on-street and off-street parking vary according to income levels, vehicle characteristics, and whether parking is paid or free [2], while motorcycle users, a dominant vehicle category in developing countries, prioritise tariff, walking distance, and queuing time over wayfinding elements [1]. Behavioural factors, including risk preferences under uncertain waiting conditions, combined with socioeconomic attributes such as income, vehicle ownership, and education, further shape parking decisions [2, 3]. Built environment characteristics, particularly road density and accessibility, also exhibit nonlinear effects on parking demand [3], confirming that parking behaviour is jointly determined by individual attributes and spatial context.

Methodologically, the parking-choice literature has employed a wide range of analytical approaches. Egan et al. [5] applied principal component analysis and cluster analysis to identify parking-type preference typologies among 574 respondents, demonstrating that demographic attributes generate distinct user clusters. Zong et al. [6] used SEM in Beijing to establish that monthly income is significantly associated with parking decisions, with higher-income users more likely to choose off-street parking and shorter parking durations. Widodo et al. [7] employed logistic regression in Sidoarjo, Indonesia, and found that age, gender, parking frequency, and parking duration are significantly associated with online parking choice, with women and younger users showing a higher propensity to adopt digital services. Zhang et al. [8] applied hierarchical regression and demonstrated that age, education, presence of children, and driving experience are significantly associated with the intention to adopt smart parking systems.

Research gap: Although parking choice behaviour has been extensively examined, existing studies concentrate on commercial districts, urban commuters, and freight operations [1–3]. The demographic heterogeneity of religious-tourism visitors, combining local motorcycle riders, family passenger-car users, individual pilgrims, and large group-tour bus passengers, and the implications of this heterogeneity for parking-zone allocation at sacred destinations remain underexplored. Moreover, few studies translate categorical demographic-preference associations directly into operational parking-zone categories that destination managers can implement.

2.2 Shared Parking and Smart Parking Systems

Shared parking has emerged as a viable response to urban parking scarcity, with growing methodological sophistication in modelling user adoption. Zhu et al. [13] combined SEM with neural networks to predict shared-parking choices in Nanjing, demonstrating that parking-location characteristics and traveller attributes jointly determine adoption. Han et al. [14] similarly applied SEM to identify latent attributes shaping shared-parking decisions, while Channamallu et al. [4] documented strong user receptivity to smart parking technologies through cluster-based analysis of parking satisfaction, recommending tiered pricing, flexible scheduling, and real-time guidance as user-satisfaction enablers. Liang et al. [12] investigated factors affecting the intention to use shared parking in Taipei City and identified perceived ease of use and trust as key behavioural drivers. Across these contributions, perceived ease of use, information availability, and user heterogeneity consistently emerge as the central design parameters for shared and smart parking systems.

Research gap: Despite methodological maturity, the shared and smart parking literature remains concentrated on urban commercial districts and university campuses [4, 12–14], with relatively homogeneous user groups such as commuters, students, and residents. The heterogeneity of religious-tourism visitors mixing local motorcycle riders, family passenger-car users, and large group-tour buses arriving in clusters around fixed worship times and the implications of this heterogeneity for shared-parking capacity allocation, bus-car spatial segregation, and the design of age-differentiated digital guidance have not been systematically addressed. Integration of shared parking principles with vehicle circulation control and peak-hour congestion management at religious destinations is, therefore, an open research area.

2.3 Transport Management at Tourism and Religious Tourism Destinations

Tourism-destination transport management has received increasing scholarly attention as visitor volumes have grown. Dewi et al. [9] analysed factors influencing mode choice at Matahari Terbit Beach, Bali, demonstrating that age and occupation are significantly associated with tourists' transport preferences. Wirasutama et al. [10] applied parking-capacity analysis at the Penglipuran Traditional Tourism Village in Bali, showing that parking-index values exceeding unity necessitate either parking-area expansion or pattern reorganisation to mitigate roadside-parking-induced congestion. Garg et al. [11] used the Analytic Hierarchy Process to prioritise motivators for visiting spiritual destinations in India, establishing that religious-tourism visitors are driven by distinct behavioural and motivational profiles relative to general tourists.

Within the religious-tourism context specifically, Pachorkar et al. [15] reviewed blockchain applications in event ticketing, crowd management, and intelligent transportation systems for large-scale religious gatherings, highlighting persistent issues of overcrowding, traffic congestion, and inefficiencies in public transport coordination during peak

pilgrimage events. Seal et al. [16] examined the role of technology, including QR-based ticketing systems, mobile applications, and crowd-monitoring tools, in enhancing visitor experience and optimising transport logistics at religious tourism destinations. Both contributions emphasise that technological interventions can substantially improve crowd and transport coordination at sacred sites, provided that the systems are tailored to the demographic profile and operational rhythm of the destination.

Research gap: Although tourism transport management has received increasing attention [9, 10], religious tourism destinations characterised by worship-time peaks, large group-tour bus arrivals, and pedestrian-dense entrance plazas remain underrepresented in the parking-management literature. Existing religious-tourism transport studies have concentrated on crowd-control technologies and digital ticketing systems [15, 16] rather than on integrated parking-management frameworks. Parking-specific solutions tailored to the operational features of religious destinations, particularly integrated frameworks combining demographic-based capacity allocation, bus-car circulation control, and age-adaptive digital parking guidance, are notably absent. The integration of demographic-based parking-preference analysis with concrete transport-management measures at religious tourism destinations is, therefore, the specific gap addressed by the present study.

3 Questionnaire Survey and Variable Design

This study translates visitor-level parking preferences into operational transport-management measures for an integrated shared parking system at a religious tourism destination. A structured quantitative approach combines questionnaire-based data collection, demographic-preference cross-tabulation, and categorical-association testing with effect-size reporting, supporting the three research questions on walking distance, digital adaptability, and vehicle-type capacity implications. Methodological choices ensure robustness to sparse-cell conditions through Fisher's Exact Test with Monte Carlo approximation and Cramer's V, while keeping variables directly interpretable as inputs to bus-car zoning, age-adaptive wayfinding, and peak-hour circulation control. The remainder of this section is organized as follows: the survey instrument and variable definitions (Section 3.1), the sampling procedure and data collection (Section 3.2), the sample characteristics (Section 3.3), and the statistical analysis (Section 3.4).

3.1 Survey Instrument and Variable Definitions

The survey was conducted on visitors to the Sheikh Zayed Grand Mosque in Surakarta who used the shared parking facilities operated as part of the area's parking-management scheme. The instrument was a structured questionnaire administered to respondents immediately after they used the parking area. The first section collected respondent demographic information (gender, age, education level, monthly income, travel purpose, and vehicle type) and the second section collected parking-preference responses. Table 1 below provides the full variable-definition table, including the question wording, response categories, measurement scale, coding for statistical analysis, and the theoretical basis for each variable.

Based on the literature synthesis, this study selected six parking-preference attributes most relevant to the Sheikh Zayed Mosque context, balancing data availability, respondent comprehension, and consistency with established parking-choice theory [17–19]. The attributes are: shared parking information system, parking availability information, parking location distance, visiting duration, parking tariff, and shared parking information comfort/simplicity. These attributes capture real-time availability needs, walking-distance tolerance, stay duration, cost sensitivity, and digital-system usability. Six demographic variables, gender, age, education level, monthly income, travel purpose, and vehicle type, were also included to test preference variation across user groups.

3.2 Sampling Procedure and Data Collection Survey Location

The survey was conducted at the shared parking facility serving the Sheikh Zayed Grand Mosque in Surakarta, Central Java, Indonesia. To capture the heterogeneity of vehicle types and user groups characteristic of religious tourism destinations, two intercept points were established within the facility: (i) the main bus and minibus (elf) parking zone, which accommodates organised pilgrim tours, and (ii) the passenger-car and motorcycle parking zone, which serves individual visitors and family groups.

Survey period and temporal coverage: Data collection took place over a four-week period and was deliberately stratified across weekdays and weekends to capture the temporal variability of parking demand. Survey shifts were aligned with the five daily prayer peaks, particularly Dhuhr, Asr, and Maghrib, and with the Friday congregational prayer, while off-peak intervals were also covered to ensure that the sample reflected the full range of arrival and dwell-time patterns observed at the facility.

Respondent selection: Respondents were intercepted immediately after parking their vehicles to ensure that responses reflected the actual parking experience rather than recalled impressions. To prevent intra-group dependence in the dataset, only one respondent per group-tour bus was surveyed, while passenger-car and motorcycle users were treated as independent observations. Participation was voluntary, and the overall refusal rate was approximately 11%, which is within acceptable limits for intercept surveys at high-traffic public-access destinations.

Table 1. Full variable definition table (questions, response categories, coding, statistical treatment, theoretical basis)

Code	Variable	Question	Response Categories	Scale/Coding	Theoretical Basis
GEN	Gender	What is your gender?	Male; Female	Nominal (1, 2)	Egan et al. [5]; Widodo et al. [7]
AGE	Age	What is your age?	<17; 17–32; 32–47; 47–52; 52–67; >67 years	Ordinal (1–6)	Zhang et al. [8]; Xue et al. [20]
INC	Monthly income	What is your monthly income?	<IDR 2 M; 2–4 M; 4–6 M; >6 M	Ordinal (1–4)	Zong et al. [6]; Yan et al. [21]
PUR	Travel purpose	What is the purpose of your visit?	Leisure/recreation; work; educational/cultural; religious; other	nominal (1–5)	Sarisoy [2]; Garg et al. [11]
VEH	Vehicle type	What type of vehicle did you use?	Motorcycle; passenger car; bus; minibus (elf)	Nominal (1–4)	Liang et al. [12]; Yan et al. [21]
SPIS	Shared parking information system (needs)	Do you feel a shared parking information system is needed at this destination?	Yes; No	Binary (1, 0)	Channamallu et al. [4]; Zhang et al. [8]
PAI	Parking availability information	Do you need real-time information on parking-space availability before/while you park?	Yes; no	Binary (1, 0)	Liang et al. [12]; Xue et al. [20]
PLD	Parking location distance (acceptable walking distance)	What walking distance from the parking facility to the mosque entrance is acceptable to you?	<100 m; 100–250 m; 250–500 m; >500 m	Ordinal (1–4)	Arunotayanun et al. [22]
VD	Visiting duration	Approximately how long do you stay at the destination per visit?	<1 h; 1–2 h; 2–3 h; 4–5 h; >5 h	Ordinal (1–5)	Zong et al. [6]; Garg et al. [11]
PT	Parking tariff (preferred/paid)	What parking fee are you willing to pay (or did you pay) per visit?	<IDR 5 k ; 5–10 k; 10–15 k; 15–20 k; >20 k	Ordinal (1–5)	Sarisoy [2]; Yan et al. [21]
SPICS	Shared parking information comfort/simplicity (perceived ease of use)	How easy is it for you to use the shared parking information system?	Easy; Not easy	Binary (1, 0)	Channamallu et al. [4]; Liang et al. [12]

Sample size: A total of 505 valid questionnaires were obtained after screening for completeness and consistency. This sample size satisfies the cell-count requirements of the categorical-association tests adopted in the analysis (Pearson Chi-Square and Fisher's Exact Test with Monte Carlo approximation) and provides adequate statistical power for the multi-level demographic cross-tabulations reported in Section 4.

3.3 Sample Characteristics

Table 2 presents the characteristics of the survey respondents. The sample comprised 235 males and 270 females. By age, 238 respondents were 17–32 years, 120 were 32–47 years, 46 were under 17, 53 were 47–52, 43 were 52–67, and 5 were above 67. By monthly income, 255 respondents earned less than IDR 2,000,000; 164 between IDR 2,000,000 and 4,000,000; 67 between IDR 4,000,000 and 6,000,000; and 19 above IDR 6,000,000. By travel purpose, 224 travelled for leisure or recreation, 206 for religious purposes, 30 for educational or cultural visits, 33 for other reasons, and 12 for work-related visits. By vehicle type, buses were the most common mode (174), followed by passenger cars (151), motorcycles (125), and minibuses (55). Most respondents indicated a preferred

walking distance of less than 100 meters from the parking facility to the mosque entrance (212 respondents), followed by 100–250 m (187), 250–500 m (70), and over 500 m (36). The duration of stay was typically 1–2 hours (227 respondents), followed by less than 1 hour (148), 2–3 hours (113), 4–5 hours (4), and over 5 hours (13). Regarding parking fees, 145 respondents paid less than IDR 5,000, 126 paid IDR 5,000–10,000, 42 paid IDR 10,000–15,000, 87 paid IDR 15,000–20,000, and 105 paid more than IDR 20,000.

Table 2. Respondent characteristics

No.	Characteristics	Frequency
1	Gender	
	Male	235
	Female	270
2	Age	
	<17 years	46
	17–32 years	238
	32–47 years	120
	47–52 years	53
	52–67 years	43
	>67 years	5
3	Income	
	<2,000,000	255
	IDR 2,000,000–IDR 4,000,000	164
	Rp 4,000,000–IDR 6,000,000	67
	>IDR 6,000,000	19
4	Travel purpose	
	Holiday/recreation	224
	Work visit	12
	Educational and cultural visit	30
	Religious visit	206
	Others	33
5	Vehicle type	
	Motorcycle	125
	Car	151
	Bus	174
	Elf	55
6	Parking location distance	
	<100 m	212
	100–250 m	187
	250–500 m	70
	>500 m	36
7	Visiting time	
	<1 hours	148
	1–2 hours	227
	2–3 hours	113
	4–5 hours	4
	>5 hours	13
8	Parking tariff	
	< IDR 5,000	145
	IDR 5,000–IDR 10,000	126
	IDR 10,000–IDR 15,000	42
	IDR 15,000–IDR 20,000	87
	>IDR 20,000	105

Several categories had small cell counts (specifically: respondents aged over 67 years, $n = 5$; work-visit respondents, $n = 12$; respondents in the highest income bracket, $n = 19$; and respondents with a stay duration of 4–5 hours, $n = 4$). Expected-cell-count diagnostics confirmed that more than 20% of the cells in several cross-tabulations had expected counts below five, violating the standard assumption of the Pearson Chi-Square Test. The Fisher's Exact Test was therefore used for all cross-tabulations affected by sparse cells, with the Monte Carlo approximation

applied to contingency tables larger than 2×2 (using 10,000 sampled tables and a 99% confidence interval for the simulated p -value). The original Chi-Square results have been re-estimated using Fisher's Exact Test, and all results are reported in Section 4.

3.4 Statistical Analysis

The categorical variables were analysed in three sequential steps. First, descriptive frequencies were computed for each demographic and parking-preference variable to characterise the sample. Second, bivariate associations between each demographic and parking-preference pair were tested for independence. Prior to testing, the expected-frequency assumption was screened for every cross-tabulation: where the assumption was satisfied, the Pearson Chi-Square Test was applied; where more than 20% of cells had an expected frequency below 5, Fisher's Exact Test was used instead. For larger tables in which exact computation was computationally intensive, Fisher's Exact Test was estimated using a Monte Carlo approximation. Of the [X] contingency tables analysed, [Y] satisfied the expected-frequency assumption and were evaluated with the Pearson Chi-Square Test, [Z] required Fisher's Exact Test, and [W] were subjected to Monte Carlo correction. Third, Cramer's V was computed as a chance-corrected effect-size measure, enabling comparison of association strength across demographic predictors. All tests were two-sided at $\alpha = 0.05$, and the findings are reported using associational rather than causal terminology, consistent with the cross-sectional and non-experimental nature of the analytical framework. These diagnostic details are reported to ensure full methodological transparency. The computational procedure is summarised in Table 3.

Table 3. Computational steps of the Chi-Square/Fisher's Exact Test procedure

Step	Description
1	Compute the expected frequencies for each cell of the cross-tabulation under the null hypothesis of independence.
2	Check the expected-frequency assumption. If >20% of cells have expected counts < 5, use Fisher's Exact Test instead of Pearson's Chi-Square (Reviewer 1).
3	Compute the Pearson Chi-Square statistic $\chi^2 = \sum ((O - E)^2 / E)$.
4	For tables larger than 2×2 in which Fisher's Exact Test is computationally expensive, use the Monte Carlo approximation with 10,000 sampled tables and a 99% confidence interval for the simulated p -value.
5	Determine the degrees of freedom $df = (\text{rows} - 1)(\text{columns} - 1)$ for Pearson Chi-Square.
6	Compare the obtained p -value with $\alpha = 0.05$ and reject the null hypothesis of independence if $p < \alpha$.
7	Compute Cramer's V = $\sqrt{(\chi^2 / (N \cdot (\min(r, c) - 1)))}$ as the effect-size measure; values < 0.10 indicate a weak association, 0.10–0.30 a moderate association, and >0.30 a strong association.

4 Results

This section reports the results of Fisher's Exact Test (with Monte Carlo approximation where applicable) for the association between five demographic variables and six parking-preference variables, together with comparative effect sizes, Cramer's V.

4.1 Gender

Gender showed statistically significant associations with two of the six parking-preference variables: visiting duration ($p < 0.001$) and parking tariff ($p = 0.001$). Female respondents reported visiting durations of 1–3 hours more frequently than males, who were more commonly present for less than one hour. Female respondents were also more likely to select higher tariff brackets (>IDR 20,000), whereas males predominantly selected the lowest tariff bracket (<IDR 5,000), suggesting a divergence in willingness to pay across genders that is consistent with longer dwell times among female visitors.

In contrast, gender showed no statistically significant association with the remaining four variables: the perceived need for a shared parking information system ($p = 0.213$), the demand for real-time parking availability information ($p = 0.105$), acceptable walking distance to the mosque entrance (parking location distance, $p = 0.970$), and the perceived ease of use of the shared parking information comfort/simplicity ($p = 0.346$). These results indicate that information-system needs, usability perceptions, and spatial walking-distance preferences do not differ meaningfully by gender at the study site. Table 4 describes gender and parking-preference variables.

Implication for parking management. The absence of significant gender-related differences across information-system and spatial-distance variables indicates that gender-targeted measures are not warranted for either the design

Table 4. Gender and parking-preference variables

No.	Parking-Preference Variable	Table Size	<i>p</i> -Value (2-Sided)	Method	Result
1	SPIS	2 × 2	0.213	Exact	Not significant
2	PAI	2 × 2	0.105	Exact	Not significant
3	PLD	4 × 2	0.970	Monte Carlo	Not significant
4	VD	5 × 2	<0.001	Monte Carlo	Significant*
5	PT	5 × 2	0.001	Monte Carlo	Significant*
6	SPICS	2 × 2	0.346	Exact	Not significant

Note: Significant associations: 2 of 6 (33.3%). Gender is associated only with visiting duration (VD) and parking tariff (PT). Shared parking information system (SPIS), parking availability information (PAI), parking location distance (PLD), and shared parking information comfort/simplicity (SPICS)

of the digital shared parking information system or the spatial allocation of parking-zone distances. However, the significant associations observed for visiting duration and parking tariff suggest two operational considerations: (i) duration-based zoning, particularly the provision of long-stay zones aligned with the 1–3-hour visiting patterns more common among female visitors during the Friday congregational prayer and family-group visits, and (ii) tariff-policy communication strategies that account for gender-differentiated willingness to pay when establishing tiered tariff structures within the shared parking system.

4.2 Age

Age was significantly associated with five of the six parking-preference variables: parking availability information ($p = 0.011$), parking location distance ($p < 0.001$), visiting duration ($p < 0.001$), T ($p < 0.001$), and shared parking information comfort/simplicity ($p = 0.022$). In contrast, shared parking information system was not significantly associated with age ($p = 0.214$), as can be seen in Table 5. Respondents aged below 32 years tended to accept shorter walking distances of less than 100 m, whereas those aged 32–47 years more frequently reported a willingness to tolerate walking distances greater than 500 m. The 47–52-year age group showed the highest proportion of respondents who did not perceive the shared parking information system as easy to use.

Table 5. Age and parking-preference variables

No.	Parking-Preference Variable	Table Size	<i>p</i> -Value (2-Sided)	Method	Result
1	SPIS	2 × 6	0.214	Monte Carlo	Not significant
2	PAI	2 × 6	0.011	Monte Carlo	Significant*
3	PLD	4 × 6	<0.001	Monte Carlo	Significant*
4	VD	5 × 6	<0.001	Monte Carlo	Significant*
5	PT	5 × 6	<0.001	Monte Carlo	Significant*
6	SPICS	2 × 6	0.022	Monte Carlo	Significant*

Note: Significant associations: 5 of 6 (83.3%). Age is associated with all preference variables except SPIS. The >67-year category ($n = 5$) was merged with the 52–67-year cohort due to expected cell counts below five. Abbreviations: SPIS = shared parking information system; PAI = parking availability information; PLD = parking location distance; VD = visiting duration; PT = parking tariff; SPICS = shared parking information comfort/simplicity; n = subgroup sample size.

Implication for parking management. These findings suggest the need for a dual-channel parking-guidance system. Digital guidance, such as mobile applications and electronic message signs, should be prioritized for the dominant 17–32-year cohort. This should be complemented by on-site human assistance and clear static signage for older visitors aged 47–67 years, particularly to address pedestrian comfort and usability needs.

4.3 Income

Income was significantly associated only with parking tariff preference ($p = 0.001$). Respondents earning less than IDR 2,000,000 showed a strong preference for the lowest tariff category, namely less than IDR 5,000. In contrast, respondents with higher income levels were more evenly distributed across the available tariff categories. No significant associations were found between income and shared parking information system ($p = 0.335$), parking availability information ($p = 0.252$), parking location distance ($p = 0.062$), visiting duration ($p = 0.085$), or shared parking information comfort/simplicity ($p = 0.440$) in Table 6.

Implication for parking management. These findings indicate that parking pricing should be sensitive to users' purchasing power. A tiered or progressive tariff structure, calibrated to local income conditions, may help maintain

equitable access for lower-income visitors while still allowing revenue generation from longer-stay users and visitors with a higher willingness to pay.

Table 6. Income and parking-preference variables

No.	Parking-Preference Variable	Table Size	<i>p</i> -Value (2-Sided)	Method	Result
1	SPIS	2 × 4	0.335	Monte Carlo	Not significant
2	PAI	2 × 4	0.252	Monte Carlo	Not significant
3	PLD	4 × 4	0.062	Monte Carlo	Not significant
4	VD	5 × 4	0.085	Monte Carlo	Not significant
5	PT	5 × 4	0.001	Monte Carlo	Significant*
6	SPICS	2 × 4	0.440	Monte Carlo	Not significant

Note: Significant associations: 1 of 6 (16.7%). Income is associated only with the parking tariff (PT). The highest income bracket ($n = 19$) was merged with the second-highest bracket, where expected frequencies fell below five. Abbreviations: SPIS = shared parking information system;

PAI = parking availability information; PLD = parking location distance; VD = visiting duration; SPICS = shared parking information comfort/simplicity; n = subgroup sample size.

4.4 Travel Purpose

Travel purpose was significantly associated with three parking-preference variables: parking location distance ($p = 0.043$), visiting duration ($p = 0.007$), and parking tariff ($p = 0.039$). Religious visitors were more likely than recreational visitors to accept longer walking distances (250–500 m), consistent with the walking element embedded in many religious practices, and tended to report longer visiting durations (1–3 hours). Work-related visitors reported the shortest visiting durations (<1 hour). Shared parking information system ($p = 0.098$), parking availability information ($p = 0.311$), and shared parking information comfort/simplicity ($p = 0.218$) were not significantly associated with travel purpose, as illustrated in Table 7.

Table 7. Travel purpose and parking-preference variables

No.	Parking-Preference Variable	Table Size	<i>p</i> -Value (2-Sided)	Method	Result
1	SPIS	2 × 5	0.098	Monte Carlo	Not significant
2	PAI	2 × 5	0.311	Monte Carlo	Not significant
3	PLD	4 × 5	0.043	Monte Carlo	Significant*
4	VD	5 × 5	0.007	Monte Carlo	Significant*
5	PT	5 × 5	0.039	Monte Carlo	Significant*
6	SPICS	2 × 5	0.218	Monte Carlo	Not significant

Note: Significant associations: 3 of 6 (50.0%). Travel purpose is associated with parking location distance (PLD), visiting duration (VD), and parking tariff (PT). The work-related visit category ($n = 12$) required Fisher's Exact Test with Monte Carlo approximation due to small subgroup size. Abbreviations: SPIS = shared parking information system; PAI = parking availability information; PLD = parking location distance; VD = visiting duration; PT = parking tariff; SPICS = shared parking information comfort/simplicity; n = subgroup sample size.

Implication for parking management. These findings suggest that parking zoning at religious tourism destinations should distinguish between short-stay zones (for work-related and quick visits, located close to the entrance) and long-stay zones (for religious and recreational visits, which can accept greater walking distances and where additional pedestrian infrastructure is justified).

4.5 Vehicle Type

In the vehicle type (Table 8), almost all aspects show vehicle type was significantly associated with five of the six parking-preference variables: shared parking information system ($p = 0.010$), parking availability information ($p = 0.003$), parking location distance ($p < 0.001$), visiting duration ($p < 0.001$), and parking tariff ($p < 0.001$). Only shared parking information comfort/simplicity ($p = 0.138$) was not significant. Bus and minibus users tolerated longer walking distances (>250 m) and reported longer visiting durations and higher tariffs, while motorcycle users dominated the short-distance (<100 m), short-duration (<1 h), and low-tariff (<IDR 5,000) brackets.

Implication for parking management. Physical separation of bus and passenger-car parking zones is essential. A dedicated bus-reservation slot system can match the strong association between vehicle type and parking-availability information needs, while a remote bus parking lot with a shuttle service to the mosque entrance is justified by the higher tolerance of bus passengers for longer walking distances.

Table 8. Vehicle type and parking-preference variables

No.	Parking-Preference Variable	Table Size	<i>p</i> -Value (2-Sided)	Method	Result
1	SPIS	2 × 4	0.010	Monte Carlo	Significant*
2	PAI	2 × 4	0.003	Monte Carlo	Significant*
3	PLD	4 × 4	<0.001	Monte Carlo	Significant*
4	VD	5 × 4	<0.001	Monte Carlo	Significant*
5	PT	5 × 4	<0.001	Monte Carlo	Significant*
6	SPICS	2 × 4	0.138	Monte Carlo	Not significant

Note: Significant associations: 5 of 6 (83.3%). Vehicle type is associated with all preference variables except SPICS. This category yielded the largest mean Cramer's V across the demographic predictors. Abbreviations: SPIS = shared parking information system; PAI = parking availability information; PLD = parking location distance; VD = visiting duration; PT = parking tariff; SPICS = shared parking information comfort/simplicity; *n* = subgroup sample size.

4.6 Comparative Effect Sizes (Cramer's V)

To allow direct comparison of association strength across demographic predictors, Cramer's V was computed for each significant cross-tabulation. The mean Cramer's V across the six parking-preference variables was largest for vehicle type, followed by age, then travel purpose, then income, and lowest for gender. Although this ordering supports the description of vehicle type as the demographic factor most consistently and strongly associated with parking-preference variation, the manuscript no longer claims that vehicle type is the "strongest determinant" in a causal or model-comparison sense. The Chi-Square/Fisher framework tests association only; causal interpretation would require a separate behavioral model.

The summary of significant associations for each parking-preference variable indicates that parking tariff is significantly associated with all five demographic factors (5/5, 100%), followed by visiting duration with four factors (gender, age, travel purpose, and vehicle type; 4/5, 80%), parking location distance with three factors (age, travel purpose, and vehicle type; 3/5, 60%), parking availability information with two factors (age and vehicle type; 2/5, 40%), and shared parking information system and shared parking information comfort/simplicity with one factor each (vehicle type and age, respectively; 1/5, 20%). The distribution of significant associations across parking-preference variables and demographic predictors is summarised in Table 9.

Table 9. Vehicle type and parking-preference variables

Demographic Factor	No. of Significant Associations	Proportion	Mean Cramer's V Rank	Interpretation
Gender	2/6	33.3%	5 (lowest)	Limited association; relevant only for VD and PT
Age	5/6	83.3%	2	Broad and strong association across most variables
Income	1/6	16.7%	4	Narrow association; relevant only for PT
Travel purpose	3/6	50.0%	3	Moderate association; relevant for PLD, VD, PT
Vehicle type	5/6	83.3%	1 (highest)	Most consistent and strongest demographic correlate

Note: Significant at $\alpha = 0.05$ ($p < 0.05$). Abbreviations: PLD = parking location distance; VD = visiting duration; PT = parking tariff.

5 Discussion

This section interprets the bivariate-association results presented in Section 4 by linking each significant statistical pattern to a concrete transport-management situation at the Sheikh Zayed Grand Mosque in Surakarta. The discussion focuses on what each demographic–preference association implies operationally for the integrated shared-parking system, including tariff structuring, vehicle-circulation control, pedestrian-flow management, and parking-guidance design. The five subsections are organised around four real-world operational situations specific to religious tourism destinations: prayer-time demand peaks, group-tour bus arrivals, elderly-visitor pedestrian flow, and digital-versus-on-site wayfinding, and conclude with a comparative positioning against previous shared-parking literature, demonstrating how the present study extends existing findings by translating categorical-association results directly into transport-engineering deliverables.

5.1 Prayer-Time Peaks and Parking Tariff

Parking tariff was significantly associated with all five demographic factors, identifying it as the most demographically heterogeneous parking-preference variable in the dataset. At a religious tourism destination, this heterogeneity interacts directly with the temporal concentration of demand around the five daily prayers, when the facility experiences sharp arrival peaks dominated by motorcycles (predominantly low-tariff users) and group-tour buses (predominantly high-tariff users). A flat tariff structure is therefore operationally inefficient. A time-of-day differentiated tariff combined with vehicle-type-specific tariff brackets would better match the underlying demand structure and support tariff-based congestion management, particularly during the Friday congregational prayer, when arrival volumes peak across all vehicle categories.

5.2 Group-Tour Bus Arrivals and Vehicle Circulation

Because vehicle type shapes nearly every parking preference, buses and minibuses (elf) should be managed as an operationally distinct user class from motorcycles and passenger cars. Accordingly, the facility should be re-zoned into (i) a bus and minibus zone with a time-slot reservation system, an off-street staging lane, and dedicated entry/exit gates to prevent queuing onto the main access road, and (ii) a passenger-car and motorcycle zone closer to the mosque entrance. Because tour buses arrive in clusters during the late-morning and early-afternoon prayer windows, the reservation system should allocate fixed arrival slots, while a marshalling area absorbs early arrivals so that available bays are not saturated within minutes.

5.3 Walking Distance for Elderly Visitors and Pedestrian Conflicts

Older visitors (47–67 years) accept shorter walking distances and find the digital parking information system harder to use. At the entrance plaza, this translates into two concrete operational requirements: a reserved short-walk parking row for elderly and disabled visitors positioned nearest the entrance, and a covered, level pedestrian walkway linking the bus zone to the plaza. During prayer-time peaks, pedestrian-priority crossings should be temporarily activated (e.g., by attendants or timed signals) to separate the elderly and family pedestrian flow from parking-exit vehicle movements.

5.4 Digital Versus On-Site Parking Guidance

Younger visitors (17–32 years), who comprised the largest share of the sample and dominated the low-tariff and short-duration brackets, showed the highest affinity for a digital parking information system, whereas elderly visitors showed the opposite pattern. A single-channel parking-guidance system is therefore inappropriate. A hybrid system is recommended: a mobile-application-based digital guidance channel for younger users, complemented by static signage, electronic message signs at the entrance, and on-site human guidance (parking attendants) for elderly visitors and large group-tour buses.

Digital parking guidance may be implemented through a combination of modalities, including: (1) mobile applications providing real-time space availability and navigational directions to designated parking zones; (2) QR-code-based directional panels installed at key entry and decision points on-site; (3) dynamic variable message signs on the primary entry access roads displaying live parking status; and (4) a real-time parking-information management system integrated with on-site traffic-control operations. These tools can serve multiple user groups: individual visitors, tour-bus operators, and site-management personnel, and may be deployed incrementally according to operational capacity and visitor volume.

5.5 Comparison with Previous Studies

The patterns observed in this study are broadly consistent with Liang et al. [12], who reported that vehicle type and perceived ease of use shape shared-parking adoption, and with the wider shared-parking literature [4, 6], which has emphasised the heterogeneity of shared-parking users. The present study extends this literature in two respects. First, this study applies shared-parking research to a new context: a religious tourism destination, where prayer schedules create sharp demand peaks and where motorcycles, cars, buses, and minibuses all share the same facility, unlike the city shopping areas and university campuses studied in previous research. Second, instead of stopping at statistical findings, this study converts the demographic-preference results into practical parking-management actions, namely separating bus and car parking zones, requiring buses to reserve slots in advance, providing different wayfinding tools for younger and older visitors, and controlling traffic flow during peak prayer times. In this way, the study delivers a transport-engineering solution, not merely a behavioural survey.

Taken together, these findings indicate that parking management during peak prayer periods should not rely on static space allocation alone. A more adaptive framework is warranted, integrating temporary redistribution of parking zones, dedicated bus-reservation areas, time-differentiated and one-way circulation protocols, and real-time digital guidance for private vehicles. Combining these measures into a single operational framework would improve both throughput efficiency and visitor experience at religious tourism sites.

6 Conclusions

This study examined how visitor demographics shape parking preferences at the Sheikh Zayed Grand Mosque, Surakarta, and translates these findings into operational measures for an integrated shared parking system at religious tourism destinations. The conclusions are organised into three parts: a statistical summary, concrete parking-management measures, and study limitations with future research directions.

6.1 Statistical Summary

Out of 30 cross-tabulations tested using Fisher's Exact Test (with Monte Carlo approximation for sparse cells), 14 (46.7%) showed significant associations at $\alpha = 0.05$. Vehicle type and age were each associated with five of the six parking-preference variables, followed by travel purpose (3), gender (2), and income (1). On the preference side, parking tariff was the most demographically heterogeneous, linked to all five demographic factors, followed by visiting duration (4), walking distance (3), availability information (2), and the shared parking information system variables (1 each). Cramer's V identified vehicle type as the strongest, though not causal, demographic correlate.

6.2 Parking-Management Measures

The findings translate into four operational measures: (1) separated bus and car zones with a slot-reservation system and dedicated entry/exit lanes for group-tour buses, preventing access-road blockages during prayer peaks; (2) tiered short-stay and long-stay zoning aligned with prayer-time peaks and group-tour arrival windows; (3) a hybrid wayfinding system combining a mobile-app-based digital channel for younger visitors supported by QR-code directional panels and variable message signs with static signage and on-site attendants for elderly visitors and large vehicles; and (4) temporary traffic and pedestrian control during peak worship periods, including reserved short-distance rows for elderly and disabled visitors and protected pedestrian crossings between the bus zone and the entrance gate.

6.3 Limitations and Future Research

Since this study focuses on a single religious tourism site, the findings should be interpreted within the specific operational context of the Sheikh Zayed Grand Mosque. Other religious tourism destinations may exhibit distinct visitor profiles, dominant travel modes (e.g., higher proportion of pedestrians or cyclists), different peak-period patterns, and varying parking demand structures. Future research should examine whether the associations identified in this study are generalisable to other categories of religious tourism sites, including those with more diverse or seasonally variable visitor flows.

Author Contributions

Conceptualization, A.M.; methodology, A.M. and N.H.; validation, G.A. and M.A.R.; investigation, G.A. and A.D.N.D.; data curation, A.D.N.D.; writing—original draft preparation, A.M.; writing—review and editing, G.A. and F.T.N.; supervision, N.H.; funding acquisition, A.M. All authors have read and agreed to the published version of the manuscript.

Funding

This work was funded by the Directorate General of Higher Education, Ministry of Education and Culture of the Republic of Indonesia, through the Regular Fundamental Research scheme of the Directorate of Research, Technology, and Community Service (DRTPM) for the 2024 fiscal year (Decree Number: 108/E5/PG.02.00.PL/2024; Agreement/Contract Numbers: 007/LL6/PB/AL.04/2024 and 196.5/A.3-III/LRI/VI/2024). Publication costs were supported by the National Muhammadiyah Research Grant Batch IX 2025 (Research Funding and Implementation Contract Number: 0259.541/I.3/D/2025).

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Acknowledgements

The authors gratefully acknowledge the Research and Innovation Institute of Universitas Muhammadiyah Surakarta; the Civil Engineering Study Program, Faculty of Engineering, Universitas Muhammadiyah Surakarta; the Surakarta City Transportation Agency; the management of the Sheikh Zayed Grand Mosque, Surakarta; the Joint Association of Parking Attendants of the Sheikh Zayed Grand Mosque, Surakarta; and all parties who contributed to the successful completion of this research.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] J. Amaya, T. Encarnación, and M. Delgado-Lindeman, "Understanding delivery drivers' parking preferences in urban freight operations," *Transp. Res. Part A Policy Pract.*, vol. 176, p. 103823, 2023. <https://doi.org/10.1016/j.tra.2023.103823>
- [2] G. Sarısoy and H. O. Tezcan, "Does parking type preference behavior differ according to whether it is paid or free? A case study in Istanbul, Türkiye," *Sustainability*, vol. 16, no. 17, p. 7269, 2024. <https://doi.org/10.3390/su16177269>
- [3] T. Peng, J. Chen, K. Liu, Z. Qiu, Z. Fu, and Y. Huang, "Examining the relationship between built environment and urban parking demand from the perspective of travelers," *J. Clean. Prod.*, vol. 385, p. 135766, 2023. <https://doi.org/10.1016/j.jclepro.2022.135766>
- [4] S. S. Channamallu, A. Pamidimukkala, S. Kermanshachi, J. M. Rosenberger, and G. Hladik, "Cluster-based analysis of parking satisfaction and strategies," *Transp. Res. Procedia*, vol. 90, pp. 790–797, 2025. <https://doi.org/10.1016/j.trpro.2025.06.078>
- [5] R. Egan, C. M. Dowling, and B. Caulfield, "Planning for diverse cycling practices: A cycle-parking type preference typology," *Case Stud. Transp. Policy*, vol. 10, no. 3, pp. 1930–1944, 2022. <https://doi.org/10.1016/j.cstp.2022.08.007>
- [6] F. Zong, P. Yu, J. Tang, and X. Sun, "Understanding parking decisions with structural equation modeling," *Phys. A Stat. Mech. its Appl.*, vol. 523, pp. 408–417, 2019. <https://doi.org/10.1016/j.physa.2019.02.038>
- [7] E. Widodo, R. N. Harnaningrum, Suparno, and A. Santoso, "Logistic regression model for user preference of online parking: A study case of Sidoarjo," *IOP Conf. Ser. Mater. Sci. Eng.*, vol. 885, no. 1, p. 012056, 2020. <https://doi.org/10.1088/1757-899X/885/1/012056>
- [8] Y. Zhang, X. Song, P. Tao, H. Li, T. Zhan, and Q. Cao, "Investigating factors for travelers' parking behavior intentions in Changchun, China, under the influence of smart parking systems," *Sustainability*, vol. 15, no. 15, p. 11685, 2023. <https://doi.org/10.3390/su151511685>
- [9] N. M. N. P. Dewi, K. V. E. M. Dwipayanti, A. Maulana, K. D. P. Suyasa, and B. Mardikawati, "Faktor yang mempengaruhi pemilihan moda transportasi wisatawan di Pantai Matahari Terbit dengan metode Chi-Kuadrat," *Berk. Forum Stud. Transp. antar Perguru. Tinggi*, vol. 1, no. 3, pp. 572–581, 2023. <https://doi.org/10.19184/berkalafstpt.v1i3.598>
- [10] C. P. Wirasutama, I. G. A. G. Suryadarmawan, and I. D. G. W. Marhaendra Putra, "Analisis kapasitas ruang parkir objek wisata desa tradisional Penglipuran Kabupaten Bangli," *J. Ilm. Kurva Tek.*, vol. 13, no. 2, pp. 190–201, 2024. <https://doi.org/10.36733/jikt.v13i2.10331>
- [11] A. Garg, P. Misra, S. Gupta, P. Goel, and M. Saleem, "Prioritizing motivators influencing intentions to visit spiritual destinations in India: An application of analytical hierarchical process (AHP) approach," *J. Tour. Futur.*, vol. 12, no. 1, pp. 38–53, 2026. <https://doi.org/10.1108/JTF-09-2021-0214>
- [12] J. K. Liang, T. Eccarius, and C. C. Lu, "Investigating factors that affect the intention to use shared parking: A case study of Taipei City," *Transp. Res. Part A Policy Pract.*, vol. 130, pp. 799–812, 2019. <https://doi.org/10.1016/j.tra.2019.10.006>
- [13] Y. Zhu, S. Chen, Y. Wu, F. Qiao, and Y. Ma, "Use of structural equation modelling and neural network to analyse shared parking choice behaviour," *Promet - Traffic & Transportation*, vol. 35, no. 5, pp. 712–721, 2023. <https://doi.org/10.7307/ptt.v35i5.209>
- [14] X. Han, D. Zhang, D. Wang, and C. Li, "Shared parking behavioral analysis based on SEM," *J. Phys. Conf. Ser.*, vol. 1972, no. 1, p. 012027, 2021. <https://doi.org/10.1088/1742-6596/1972/1/012027>
- [15] P. R. Pachorkar, V. B. Ghuge, R. R. Antapurkar, V. S. Gaikwad, V. L. Bhusari, and N. Ujgare, "A comprehensive survey of blockchain applications in event ticketing, crowd management, and intelligent transportation systems for large-scale religious gatherings," in *2025 International Conference on Future Technologies (ICFT)*, 2025, pp. 1–6. <https://doi.org/10.1109/ICFT66708.2025.11336404>
- [16] M. Seal, P. Singh, and D. Sharma, "Transforming visitors' experiences and site management: Role of technology in enhancing services at religious tourism destinations," in *Technology and Religious Tourism: Emerging Trends, Cases and Futuristic Perspectives*. Emerald Publishing Limited, 2025, pp. 49–60. <https://doi.org/10.1108/978-1-83662-260-420251004>
- [17] M. F. Jonker, "Level overlap and level color coding revisited: Improved attribute attendance and higher choice consistency in discrete choice experiments," *J. Choice Model.*, vol. 52, p. 100494, 2024. <https://doi.org/10.1016/j.jocm.2024.100494>

- [18] A. L. S. de Sá and C. S. Pitombo, “Methodological proposal for stated preference scenarios regarding an exploratory evaluation of ride-hailing implications on transit: A Brazilian context analysis,” *Case Stud. Transp. Policy*, vol. 9, no. 4, pp. 1727–1736, 2021. <https://doi.org/10.1016/j.cstp.2021.07.020>
- [19] V. Díaz, R. Montoya, and S. Maldonado, “Preference estimation under bounded rationality: Identification of attribute non-attendance in stated-choice data using a support vector machines approach,” *Eur. J. Oper. Res.*, vol. 304, no. 2, pp. 797–812, 2023. <https://doi.org/10.1016/j.ejor.2022.04.018>
- [20] Y. Xue, Q. Kong, F. Sun, M. Zhong, H. Tu, C. Li, and H. Guan, “Shared parking decision behavior of parking space owners and car travelers based on prospect theory—A case study of Nanchang City, China,” *Sustainability*, vol. 14, no. 24, p. 16877, 2022. <https://doi.org/10.3390/su142416877>
- [21] Q. Yan, T. Feng, and H. Timmermans, “Private owners’ propensity to engage in shared parking schemes under uncertainty: Comparison of alternate hybrid expected utility-regret-rejoice choice models,” *Transp. Lett.*, vol. 15, no. 7, pp. 754–764, 2023. <https://doi.org/10.1080/19427867.2022.2088568>
- [22] K. Arunotayanun, P. Sukpattanachoen, S. Manokeaw, P. Khankham, and D. Rinchumphu, “Decision factors to walk from house to a park in housing estate projects: Case of Chiang Mai, Thailand,” in *Proceedings of the 11th World Congress on New Technologies (NewTech’25)*, Paris, France, 2025, p. ICCEIA 120. <https://doi.org/10.11159/icceia25.120>

Nomenclature

V	Cramer’s V (chance-corrected effect-size measure for categorical associations)
p	Probability value (two-sided) of the statistical test
N	Total sample size (number of valid questionnaires)
O	Observed frequency in a contingency table cell
E	Expected frequency in a contingency table cell under the null hypothesis of independence
r	Number of rows in a contingency table
c	Number of columns in a contingency table
df	Degrees of freedom, $df = (r - 1)(c - 1)$
PT	Parking tariff (preferred or paid per visit)
h	Hours (unit of visiting duration)
m	Metres (unit of walking distance)
SPIS	Shared parking information system (needs)
PAI	Parking availability information
PLD	Parking location distance (acceptable walking distance)
VD	Visiting duration
PT	Parking tariff (preferred/paid)
SPICS	Shared parking information comfort/simplicity (perceived ease of use)
GEN	Gender
AGE	Age cohort
INC	Monthly income
PUR	Travel purpose
VEH	Vehicle type (motorcycle, passenger car, bus, minibus/elf)
CI	Confidence Interval
df	Degrees of freedom
IDR	Indonesian Rupiah
ITS	Intelligent Transport System
MC	Monte Carlo (simulation/approximation)
RQ	Research question
SEM	Structural Equation Modeling
SP	Shared parking
(2×2)	Two-by-two contingency table (the smallest cross-tabulation in this study)

Greek symbols

α	Significance level for hypothesis testing ($\alpha = 0.05$)
χ^2	Pearson Chi-Square Test statistic, $\chi^2 = \sum \frac{(O-E)^2}{E}$
$\sum$	Summation operator across contingency-table cells
$\sqrt{\cdot}$	Square-root operator (used in the Cramer’s V formula)

Subscripts and modifiers

i	Cell index within a contingency table (row–column position)
$\min(r, c)$	Minimum of the row and column counts in a contingency table (used in Cramer’s V denominator)
2-sided	Two-sided p -value, reported for all bivariate association tests
exp.	Expected (as in expected cell frequency, E)
obs.	Observed (as in observed cell frequency, O)
MC	Monte Carlo approximation (applied to Fisher’s Exact Test for tables larger than (2×2))